Paul MacLean

paul-maclean.com 412-389-7756 macleanp98@gmail.com Pittsburgh, PA

Education

MS in Computer Science, West Virginia University, Morgantown, WV

BS in Computer Science, California University of Pennsylvania, California, PA

Experience

Agile 5 Technologies **November 2022 – November 2025**

Modernization: Full Stack Developer

- Architected a modernized SSR frontend for regulations.gov using SvelteKit and TailwindCSS
- Translated UX wireframes and product owner requirements into NestJS endpoints and SvelteKit components
- Integrated frontend and backend OIDC authentication with Login.gov
- Ensured product was tested, branded, compliant, and secure

Legacy: Frontend Developer

- Deployed and maintained legacy Ember and JS Spring Boot microservice systems for regulations.gov
- Spearheaded new comment clustering featureset, halving review time and saving thousands of manhours

Innovation Lab: Full Stack Developer

- Designed and implemented a rule engine that assesses NIST 800-53 cloud compliance across via a NextJS/Pocketbase monolith
- Added security scanning to CI/CD pipeline; uploaded findings to CVE tracking warehouse

West Virginia University **August 2020 – May 2022**

Graduate Research Assistant

- Trained thousands of models to predict programmer effort using Jira and GitHub data for NASA IV&V
- Engineered bespoke Python pipeline to mine, store, structure, and classify data from open-source repos
- Graded Analysis of Algorithms fundamentals (data structures, complexity, sorting, graph algorithms)

U.S. Steel **May 2019 – September 2019**

Process Control Engineer (Internship)

- Learned the inner workings of the steelmaking process, and the integration of technology throughout
- Fixed bugs in a VMS operating system, Fortran, and SQL stack

Skills

- **Tools and Concepts:** NodeJS, NestJS, NextJS, SvelteKit, Vitest, Jest, Tailwind, Spring, Microservices, AWS, Git, BitBucket, Machine Learning, Playwright, CI/CD, EmberJS, Agile/Scrum
- **Languages:** TypeScript, Python, Java, C, C++, Lua
- **Consulting & Collaboration:** System Thinking, Solution Design, Client Engagement, Troubleshooting, Knowledge Transfer, Jira, Confluence

Publications

Paul MacLean, Tom Devine, Dale Dzielski, Joshua Hernandez (2023). Efficacy of Reported Issue Times as a Means for Effort Estimation. CSCE 2023, IEEE. ISBN: 979-8-3503-2759-5

Projects

Othello Boardgame Digitizer

Trained neural network to classify Othello tiles as empty, white, or black

Applied image processing techniques to transform off-angle board images into 2D arrays for virtual play

Vulcan Transit Tracker

Built an IoT prototype for an Arduino bus tracking system for alma mater shuttle serviceInterfaced with 4G module via AT commands to send coordinates to Spring backend